

Maximising the coverage of emergency medical services with time-dependent demand restrictions: a case study in Shanghai

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ABSTRACT

In many Asian cities, periodic changes in the demand for emergency medical services are concerned, but few existing studies have discussed the time-dependent factor in delivering these services. This study proposes a formula for locating emergency medical services, in which factors of varying demand, time-dependence, cost and facility capacity are considered.

A real case is investigated in order to understand the impact of time-dependence on decision making. The case study suggests different solutions depending on whether the design is service-oriented or cost-oriented.

The analysis also indicates that the limitations on demand complicate the making of decisions to optimise service locations, e.g. more service stations are required. By creating a design for allocating service sites based on demand nodes in a real case, this study provides a framework for analysing the location problem when demand changes.

KEYWORDS Emergency medical services; integer programming; location; time-dependent restrictions

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1. Introduction

The Shanghai society faces a variety of challenges due to rapid urbanisation. At the end of 2014, the elderly population in Shanghai was 28.8% of the total population (Shanghai Civil Affairs Bureau, 2015). The growing proportion of elderly residents, together with the booming number of migrants is placing growing pressure on the city's emergency medical services. A significant growth in

staff workloads can be easily detected, as indicated below in the summary of emergency medical services in Shanghai from 2001 to 2014. As a result, the response time for ambulance services in urban cities is getting longer (Qi and Jing, 2017).

Along with the increasing demand for medical services, other problems have emerged during the implementation of a more effective service, such as limited resources and changes in demand. This paper observed that the time-dependent

Table 1. Summary of emergency medical services in Shanghai. (source: www.sn120.sh.cn)

Year	Distance travelled (km)	Number of vehicle-times	Number of patients
2001	3,684,900	157,400	146,600
2002	4,757,000	192,200	179,800
2003	5,540,600	223,200	207,500
2004	6,237,800	254,000	234,200
2005	6,898,700	297,400	268,300
2006	7,573,300	327,200	290,100
2007	8,251,600	359,000	322,800
2008	9,052,700	396,800	361,900
2009	9,632,700	430,200	396,100
2010	11,330,100	498,700	455,800
2011	-	535,900	487,800
2012	12,016,600	594,500	541,000
2013	12,648,300	628,500	571,900
2014	12,367,700	642,000	584,000

aspect of the demand problem is mainly being caused by the mobility of citizens.

To put it simple, a large portion of the population is now choosing to live in the growing suburbs and commute downtown to work. Thus, the movement of the population has substantially changed the demand for emergency services in different regions of the city at different times. Where one medical station might have once been sufficient for communities in a designated region, the fluctuating overall demand for ambulances from a shifting population often leads to long response times by medical services.

These shifting population patterns also influence the planning procedures for emergency facility locations. So, optimising facility locations for emergency medical services has become a priority for local government.

This study is driven by the time-dependent nature of the demand for emergency medical services. Shackled by limited budgets and finite land resources, planners have to consider many factors such as the city layout, costs, the limited number of candidate sites and the mobility of the population, in order to optimise the use of facilities while achieving a satisfactory service level. Consequently, the allocation problem for emergency medical sites needs to be investigated from a variety of different angles to ensure the optimum allocation of facilities for each demand node. In this research, a real case in Shanghai is discussed.

2. Literature review

The importance of ambulance services in any urban society has been explored extensively in the literature. Many problems are considered such as the location of ambulance stations, and the relocation and dispatching of ambulances.

These problems are connected to each other, but solving the facility location problem is the basis for making all the other related decisions (Aringhieri et al., 2017).

Minimising the cost of covering all the demand nodes (Toregas et al., 1971) and covering the largest possible level of demand (Church and ReVelle, 1974) are some recognised objectives of previous research. Similar objectives have appeared in studies on determining the locations of fire stations (Indriasari et al., 2010) and emergency warning sirens (Current and O'Kelly, 1992).

Early research work, however, has some major drawbacks, e.g. the assumption of un-capacitated facilities and the static condition of demand, largely negate the application of current coverage models.

To remedy this situation, more attention must be paid to improve the models to reflect real-world problems. Hence, some researchers (Hogan and ReVelle 1986; Gendreau et al., 1997) considered the Double Standard Model to ensure multiple coverage of each demand node, while other researchers (Daskin 1983; Batta et al., 1989) extended the studies on investigating the availability of ambulances at different time periods. More recently, the

dynamic setting of ambulance locations (Gendreau et al., 2001; Rajagopalan et al., 2008; Zarandi et al., 2013) was considered. Another direction for applying current coverage models to real problems is to consider capacitated facilities (Chung et al., 1983; Current and Storbeck, 1988; Murray and Gerrard, 1997). Applications of the extended capacitated covering models to real-world problems are provided by authors (Yin and Mu, 2012).

In recent years, the time-dependent aspect in coverage problems has attracted growing attention. One of the features is that the speed of the ambulance is affected by unfavourable traffic conditions during rush hours, and therefore the radius of prompt cover can vary at different times (see arguments on vehicle routing problems in Hill and Benton, 1992; Malandraki and Daskin, 1992; Ichoua et al., 2003).

This led to the problems of ambulance location and relocation in relation to time-dependent service (Budge et al., 2010; Schmid and Doerner, 2010). In medical emergency logistics, the time window was also an important factor when optimising the number and locations of supply nodes to demand nodes (see a case study by Tuzkaya et al., 2014).

Some other researchers (Griffin et al., 2008) argued that the cost of health services and the locations of health centres are subject to fluctuations in needs, while some recent surveys (Başçar et al., 2012; Farahani et al., 2012) highlighted the fact that the time-dependence aspect of emergency service station location problems were under-developed areas of academic research.

On the one hand, rapid urbanisation in Asia and an ageing population have resulted in a growing need for more ambulances and health care services overall, (Loo and Lam, 2012), while the residential shifts of people has exacerbated the delivery of the services. On the other hand, however, we can see from the literature that few studies have discussed models and applications for covering problems of delivering medical services with time-dependent demand restrictions.

Therefore, on the basis of some real cases in Asian cities, a formula addressing two time-dependent demand constraints was considered in this research. A better plan for the location of medical stations was attempted with reference to changing demands during different time periods.

3. Model formulations

More background on the planning process is introduced to provide the mathematical formulation. In reality, a new plan for facilities is often made from the existing service locations and some new potential candidates. The budget restricts the number of locations in this plan, and also imposes an upper limit on the capacity of medical emergency services at each site.

In order to facilitate the service process and control the costs, one demand node is assigned to at most one service

site. The demands of a node correspond to the population, while the demand varies periodically over time. More specifically, it is also affected by work time and leisure time. Authorities will always attempt to improve service levels, and therefore, in this research, maximising the coverage of emergency services within a limited budget, and time-dependent restriction were the major objectives.

The problem is formulated as follows:

Parameters:

- N the set of existing and candidate emergency service sites,
- N^0 the set of existing emergency service sites at the beginning of the plan, i.e. $N^0 = \{j \in N \mid x_j^0 = 1\}$ where x_j^0 denotes the initial status of site j ,
- M the set of demand nodes,
- $a_{ij} = \begin{cases} 1, & \text{if the distance from } i \text{ to } j \text{ is not greater than the} \\ 0, & \text{upper limit of services,} \\ & \text{otherwise} \end{cases}$
- f_j the cost of setting up emergency service at site j or removing the service from j ,
- v_j the cost of maintaining emergency services at site j ,
- TC the upper limit of costs for setting up emergency services at candidate sites and removing emergency services from existing facilities,
- h_i the population of a demand node i in working hours,
- h'_i the population of a demand node i in non-working hours,
- C_j the capacity of emergency services at site j .

Variables:

- $x_j = \begin{cases} 1, & \text{if emergence services are positioned at site } j, \\ 0, & \text{otherwise.} \end{cases}$
- $z_{ij} = \begin{cases} 1, & \text{if demand node } i \text{ is services by site } j, \\ 0, & \text{otherwise.} \end{cases}$

The integer programming formulation is as follows:

$$\text{Max}(\sum_{i \in M} \sum_{j \in N} \alpha h_i z_{ij} + \sum_{i \in M} \sum_{j \in N} (1 - \alpha) h'_i z_{ij}) - w(\sum_{j \in N} v_j x_j + \sum_{j \in N^0} f_j (x_j^0 - x_j) + \sum_{j \in N \setminus N^0} f_j x_j),$$

Subject to:

$$\sum_{j \in N} z_{ij} \leq 1 \quad \forall i \in M, \quad (1)$$

$$z_{ij} \leq \min \{a_{ij}, x_j\} \quad \forall i \in M, j \in N, \quad (2)$$

$$\sum_{i \in M} a_{ij} h_i z_{ij} \leq C_j x_j \quad \forall j \in N, \quad (3)$$

$$\sum_{i \in M} a_{ij} h'_i z_{ij} \leq C_j x_j \quad \forall j \in N, \quad (4)$$

$$0 \leq \sum_{j \in N^0} f_j (x_j^0 - x_j) + \sum_{j \in N \setminus N^0} f_j x_j \leq TC, \quad (5)$$

$$x_j \in \{0, 1\} \quad \forall j \in N, \quad (6)$$

$$z_{ij} \in \{0, 1\} \quad \forall i \in M, j \in N. \quad (7)$$

The objective of this model is to maximise the coverage of emergency services, i.e. $\sum_{i \in M} \sum_{j \in N} \alpha h_i z_{ij} + \sum_{i \in M} \sum_{j \in N} (1 - \alpha) h'_i z_{ij}$, while minimising the total cost of providing the services, i.e. $\sum_{j \in N} v_j x_j + \sum_{j \in N^0} f_j (x_j^0 - x_j) + \sum_{j \in N \setminus N^0} f_j x_j$.

The service level is measured by a weighted summation of the population, e.g. α is the weight for the population of demand nodes during working hours. In order to analyse the impact of the objective function on solutions, w as a weight to the second part of the objective was introduced.

Constraint (1) ensures that each demand node is serviced by at most one emergency service site. Constraint (2) states that a demand node i cannot be serviced by a site j , when the node is not covered by the service of j or the emergency service is not positioned at j .

Constraints (3) and (4) ensure that the assignment of demands i to j cannot exceed the capacity of j . Constraint (5) gives the budget limitations on setting up new emergency service sites and removing the service from some existing sites.

Constraints (6) and (7) are the integrality constraints.

In the above mixed integer and linear programming formulation, Constraints (3) and (4) are introduced in consideration of the time-dependent feature of demand. Constraint (3) refers to the population which could benefit from the emergency medical services during business hours.

Constraint (4) considers the number of citizens which could be serviced during leisure time. These two constraints, together with the objective function, achieve the goal of maximising the overall service level under budget restrictions.

Noting that a simple set coverage problem is a classic NP-complete problem (Gary and Johnson, 1979). In the literature, various algorithms and heuristics have been discussed (Farahani et al., 2012), but optimal solutions can hardly be ensured. Some researchers (Sorensen and Church, 2010) exhibited good results by using the ILOG CPLEX mixed-integer programme solver on solving real cases of specific set coverage problems. Hence, the commercial mathematical solver in experiments was attempted, and it returned satisfactory and feasible solutions to the problem instances of a real case (see details in Section 3).

4. Application

In this section, a real case in Shanghai, a city of China, is discussed. As mentioned previously, the growing population and an expanding boundary of the city is placing growing pressure on the city government to satisfy the changing needs for emergency medical services.

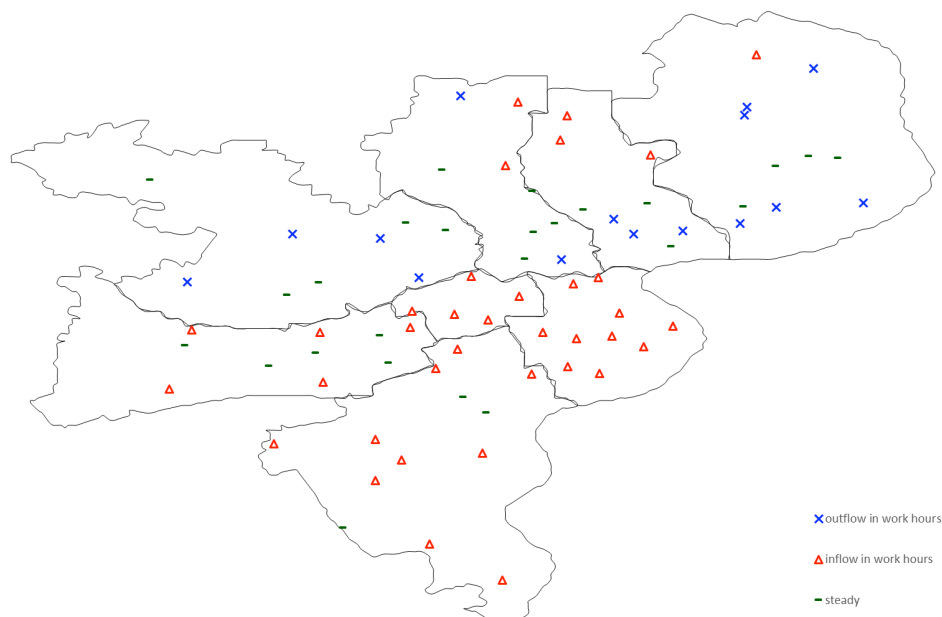


Figure 1. The change of population in sub-districts.

The city government attempted to establish more ambulance stations to improve the service, in particular, in the central region of the city, because it accommodates more than one-third of the population, and provides a majority of job opportunities. Existing facilities include 33 emergency medical sites, which are distributed across eight districts of this central region of Shanghai.

In order to reflect a real problem, great effort was put into investigating the census, i.e. the demographic yearbooks and reports released by the city medical services authority. Parameter values in this case study are given on the basis of the investigation.

4.1. Data input

The eight districts in the central region of Shanghai consist of 76 sub-districts, and these sub-districts are regarded as demand nodes in this study. Noting that the Chinese government conducts the population census every 10 years, and therefore the most completed data for the population is updated only till 2011. The demographic yearbook 2011 in Shanghai districts (Statistics Department, 2011a-2011h) was referred, and data were input to the parameters of h_i . The values of parameter h_i were estimated by referring to the report on the occupational composition of the employed population in Shanghai districts (Statistics Department, 2009). According to these values of h_i and h_i' , a figure for the population change in sub-districts is outlined to demonstrate the major feature of the periodic change in demand.

As shown in Figure 1, a net outflow of the population in working hours is represented by crosses, while triangles

give a net inflow of the population during working hours. It is clear from the figure that the population stays relatively steady in a few sub-districts which is denoted by bars.

Noting that the working hours defined in this study include business hours, i.e. from 9:00am to 5:00pm, and some transit time between the workplace and living place. It is common in Shanghai that an employee needs to spend around two or three hours on commutation each day. In other words, he/she spends approximate one-third of his/her time on work in each week. Hence, without loss of generality, $\alpha = \frac{1}{3}$ was set.

When deciding the value of N , another common practice that ambulances are often stationed in hospitals or medical centres in Chinese cities was noted. Hence, after investigating all possible locations to set up new emergency medical sites in central Shanghai, there were 146 candidates (Appendix 1). In other words, $N^0 = \{1, \dots, 33\}$, and $N = \{1, \dots, 33, 34, \dots, 179\}$.

The values of parameters were determined by referring to the 12th five-year plan on the development of Shanghai emergency medical services (Shanghai Municipal Commission of Health and Family Planning, 2013). In this plan, the standard for the covering radius of each site is five kilometers, and thus the value of a_{ij} was determined. By referring to the population in the central region and the standard of the ambulance service, the capacity of a service site should cover a population of no more than 250,000, i.e. $C_j = 25$ was set for each j . According to the criteria for setting up emergency medical sites in Shanghai (Statistics Department, 2003), the cost for building a new site is between one million and 1.5 million, while the operational

cost is one quarter. Hence, without loss of generality, let $f_j = 10$ for $j \in N^0$, and $f_j = 100$ for $j \in N/N^0$, and let $v_j = 25$ for each $j \in N$. In the 12th five-year plan (Shanghai Municipal Commission of Health and Family Planning, 2013), the authority would establish or reconstruct five or less emergency medical facilities in the central region of Shanghai. Thus, the budget limit TC is 500.

4.2. Results and Discussions

In order to analyse this problem more comprehensively, four instances with different settings were tested, i.e. $w = 0.1$, $\alpha = \frac{1}{3}$ with Constraints (1) - (7), $w = 0.1$, $\alpha = 0$ with Constraints (1) - (2) and (4) - (7), $w = 1$, $\alpha = \frac{1}{3}$ with Constraints (1) - (7), and $w = 1$, $\alpha = 0$ with Constraints (1) - (2) and (4) - (7).

More specifically, the setting $w = 0.1$ implies that the design in Instance I and II is service-oriented, but changes in demand would not be considered in Instance II. In contrast, the setting $w = 1$ suggests that the factor of costs will be more weighted in the design. Instance III and IV, and Instance IV will not consider the daytime restriction on demand, i.e. Constraint (3).

All instances are tested by IBM ILOG CPLEX 12.4 running on a desktop computer with a 3 GHz processor. A limit of one hour is set on each computation. CPLEX returned an optimal or a satisfactory solution to Instance II in 64.5 seconds. It returned satisfactory solutions to Instances I, III and IV in 3,600 seconds. The solution gap is 1.6%, 0%, 8.9% and 4.7%, respectively. More detailed results are presented and demonstrated below.

First, the results of selected emergency medical stations were depicted on the following figures.

When deciding the values of w , the difference in results with different practical focuses were attempted to explore. In fact, it is implied from $w = 0.1$ that the government concerns more with the service level, i.e.

$$\sum_{i \in M} \sum_{j \in N} \alpha h_i z_{ij} + \sum_{i \in M} \sum_{j \in N} (1 - \alpha) h_i' z_{ij}, \text{ but not the cost.}$$

In contrast, the authority has to accept a more economical solution for locations when this weight is large, i.e. $w = 1$. Therefore, it is indicated from Figures 2 and 3 that the authority would set up 37 and 33 locations for emergency medical services when $w = 0.1$, while it only needs 33 and 32 sites when the budget is more important when $w = 1$. Furthermore, two sub-figures were compared, i.e. with and without periodic changes in demand, in each of Figures 2 and 3.

The results reveal another critical fact that varying demand complicates the planning problem, as the number of selected locations for I and III are greater than those for II and IV. Under similar situation in two settings of w , the authorities need to set up more stations to satisfy the time-

dependent demands of ambulance services. It is worth mentioning the results in II and III that the solutions are exactly the same with existing emergency service sites in N^0 . That is to say, the existing facilities in the central region of Shanghai can meet the requirements for such services if the population would not vary periodically, or if the authorities have to consider budget constraints. They could even remove one site from these existing sites in the setting of IV, i.e. the cross as shown in Figure 3.

Next, the results from the prospect of a covering level were explored, i.e.

$$\frac{\sum_{i \in M} \sum_{j \in N} \alpha h_i z_{ij} + \sum_{i \in M} \sum_{j \in N} (1 - \alpha) h_i' z_{ij}}{\sum_{i \in M} \alpha h_i + \sum_{i \in M} (1 - \alpha) h_i'}.$$

The results of the covering level for four instances, i.e. I, II, III and IV, are 100%, 100%, 91.99% and 98.61%, respectively. That is to say, all demands are covered by the 37 service sites in Instance I and by 33 sites in Instance II. The authorities reduce the number of locations in Instances III and IV, but at the cost of service level, i.e. 91.99% and 98.61%. It is interesting to note that the covering level of IV is greater than that of III, even with fewer service sites. More detailed results of the allocation of facilities to demand nodes are presented as follows.

The detailed allocation pattern for the four instances given in Table 2 was compared, and thus more properties for this realistic problem could be explored.

Under the same settings to w and other parameters, the allocation of facilities to demand nodes in I and III is quite different to the result of II and IV, respectively. This result implies that the allocation pattern of service sites to demand nodes needs to be rearranged when attempting a satisfactory covering level, but with periodic changes in demand.

5. Concluding remarks

In this research, the location problem for emergency medical services with time-dependent demand restrictions was investigated. Multiple factors in the decision making process were considered, i.e. periodic changes in demand, costs and the capacity of the facilities. Time-dependent demand restrictions into the formula were introduced, and maximising the coverage of emergency services while minimising the total cost of providing the services were aimed.

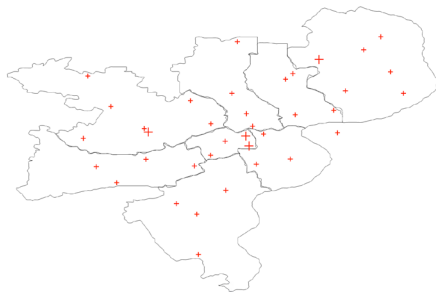
The formula is used to study a real problem in Shanghai. The case is discussed with different parameter settings, for example, when the design is more service-oriented or cost-oriented, and whether the time-dependent restrictions exist or not.

Some interesting conclusions are reached from the analysis. Firstly, both the weight in the objective function and the time-dependent constraints change the result of locations and allocation of service sites to demand nodes.

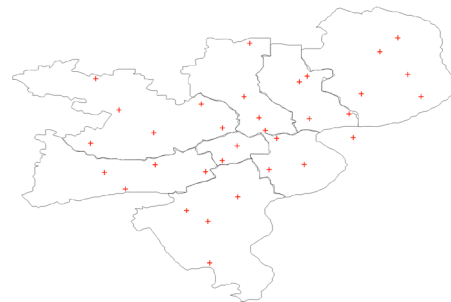
Table 2. The allocation of facilities to demand nodes.

Demand Facility Instance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
I	27	16	3	8	24	4	33	2	7	25	1	18	19	5	15
II	24	11	24	29	33	8	22	11	2	4	3	28	5	19	18
III	11	4	8	23	7	16	15	24	33	22	18	6	1	18	2
IV	11	2	24	11	4		6	1	33	5	18	17	25	2	6
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
I	12	1	14	20	25	18	29	24	13	103	20	6	5	32	124
II	12	17	20	5	19	28	18	19	20	18	20	14	31	13	27
III	28	19		20	24	17	5		4		17	31	13	32	31
IV	27	19	12	12	7	20	29	17	17	29	18	6	14	13	31
	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
I	124	124	31	28	17	30	30	15	17	31	13	22	14	19	28
II	12	14	32	7	6	12	6	30	4	14	13	3	17	25	1
III	31		14		27	28	25		13	20	6	19	32	27	1
IV	28	31	32	3	28	7	5	33	22	20	28	24	14	19	13
	48	46	47	49	50	51	52	53	54	55	56	57	58	59	60
I	6	32	12	4	103	118	6	27	33	172	118	29	23	21	11
II	2	31	32	33	27	6	1	27	22	5	25	23	16	7	30
III	30	12	14	16	7	30	11		25	30	12	29	2	3	29
IV	30	31	32	1	30	5	29	30	27	7	12	23	8	3	25
	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
I	10	22	23	7	16	9	2	3	9	26	8	11	172	10	26
II	10	29	30	23	21	21	2	8	15	26	16	26	9	10	9
III	10	5	15	22	23	26	26	8	10	21	9	9		33	21
IV	22	4	25	15	9	16	33	16	23	10	8	10	21	15	21
	76														
I	26														
II	15														
III	21														
IV	9														

(a)



(b)

Figure 2. Results of locations of emergency medical sites when $w = 0.1$. (a) with periodic changes in demands; and (b) without periodic changes in demands.

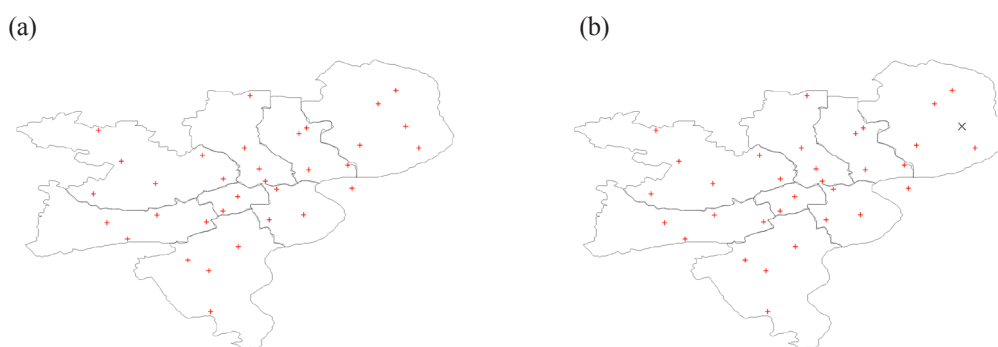


Figure 3. Results of locations of emergency medical sites when $w = 1$. (a) with periodic changes in demand; and (b) without periodic changes in demand.

Furthermore, the time-dependent constraints could affect the final decision on the locations. More service sites are required to satisfy varying demands, but the service coverage level is still unfavourable in one test instance.

This study highlights a specific focus on the location problem with time-dependent aspects. The mobility of the population has put growing pressure on medical services in many cities in China which are going through the mega changes of rapid urbanisation.

This study does not consider more features, e.g. some locations may have higher accident rates than others. But the design shows a framework for optimising facility locations for emergency medical services in cities and a basis for relocating and dispatching ambulances. Hence, further research work can be conducted in future, for instance, when the change of demand appears to be more dynamic or when more features are considered.

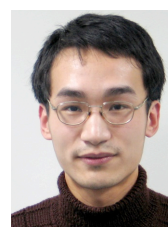
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References

- [1] Aringhieri R, Bruni ME, Khodaparasti S and Essen JTV (2017). Emergency medical services and beyond: Addressing new challenges through a wide literature review. *Computers and Operations Research*, Volume 78 (C), pp. 349-368.
- [2] Başçar A, Çatay B and Ünlüyurt T (2012). A taxonomy for emergency service station location problem. *Optimization Letters*, 6(6), pp. 1147-1160.
- [3] Batta R, Dolan JM and Krishnamurthy NN (1989). The maximal expected covering location problem: revisited. *Transportation Science*, 23(4), pp. 277-287.
- [4] Budge S, Ingolfsson A and Zerom D (2010). Empirical analysis of ambulance travel times: The case of calgary emergency medical services. *Management Science*, 56(4), pp. 716-723.
- [5] Chung C, Schilling D and Carbone R (1983). *The capacitated maximal covering problem: A heuristic*. In: Proceedings of the fourteenth annual Pittsburgh conference on modeling and simulation. Pittsburgh, pp. 1423-1428.
- [6] Church R and ReVelle C (1974). The maximal covering location problem. *Papers in Regional Science*, 32(1), pp. 101-118.

- [7] Current JR and O'Kelly ME (1992). Locating emergency warning sirens. *Decision Sciences*, 23(1), pp. 221-234.
- [8] Current JR and Storbeck JE (1988). Capacitated covering models. *Environment and Planning B: Planning and Design*, 15, pp. 153-164.
- [9] Daskin MS (1983). A maximum expected covering location model: Formulation, properties and heuristic solution. *Transportation Science*, 17(1), pp. 48-70.
- [10] Farahani RZ, Asgari N, Heidari N, Hosseini M and Goh M (2012). Covering problems in facility location: A review. *Computers & Industrial Engineering*, 62(1), pp. 368-407.
- [11] Garey MR and Johnson DS (1979). *Computers and intractability: A guide to the theory of NP-completeness*. New York: Freeman.
- [12] Gendreau M, Laporte G and Semet F (1997). Solving an ambulance location model by tabu search. *Location Science*, 5(2), pp. 75-88.
- [13] Gendreau M, Laporte G and Semet F (2001). A dynamic model and parallel tabu search heuristic for real-time ambulance relocation. *Parallel Computing*, Volume 27, pp. 1641-1653.
- [14] Griffin PM, Scherrer CR and Swann JL (2008). Optimization of community health center locations and service offerings with statistical need estimation. *IIE Transactions*, 40(9), pp. 880-892.
- [15] Hill AV and Benton WC (1992). Modelling intra-city time-dependent travel speeds for vehicle scheduling problems. *The Journal of the Operational Research Society*, 43(4), pp. 343-351.
- [16] Hogan K and ReVelle C (1986). Concept and applications of backup coverage. *Management Science*, 32, pp. 1434-1444.
- [17] Ichoua S, Gendreau M and Potvin JY (2003). Vehicle dispatching with time-dependent travel times. *European Journal of Operational Research*, 144(2), pp. 379-396.
- [18] Indriasari V, Mahmud AR, Ahmad N and Shariff ARM (2010). Maximal service area problem for optimal siting of emergency facilities. *International Journal of Geographical Information Science*, Volume 24(2), pp. 213-230.
- [19] Loo PY and Lam WY (2012). Geographic accessibility around health care facilities for elderly residents in Hong Kong: A microscale walkability assessment. *Environment and Planning B: Planning and Design*, 39, pp. 629-646.
- [20] Malandraki C and Daskin MS (1992). Time-dependent vehicle routing problems: Formulations, properties and heuristic algorithms. *Transportation Science*, 26(3), pp. 185-200.
- [21] Murray AT and Gerrard RA (1997). Capacitated service and regional constraints in location-allocation modeling. *Location Science*, 5(2), pp. 103-118.
- [22] Qi TF and Jing J (2017). Response time of pre-hospital first aid in urban regions in China, 1996-2015: A meta-analysis. *Chinese Journal of Public Health*, 33(10), pp. 1466-1468.
- [23] Rajagopalan HK, Saydam C and Xiao J (2008). A multi-period set covering location model for dynamic redeployment of ambulances. *Computers & Operations Research*, 35(3), pp. 814-826.
- [24] Schmid V and Doerner KF (2010). Ambulance location and relocation problems with time-dependent travel times. *European Journal of Operational Research*, 207(3), pp. 1293-1303.
- [25] Shanghai Civil Affairs Bureau (2015). *2014 report of elderly population and public affairs*. Shanghai: Shanghai Civil Affairs Bureau.
- [26] Shanghai Health and Family Planning Commission. *The twelfth 5-years plan on the development of Shanghai emergency medical services*. [online]. Available at <<http://www.wsjsw.gov.cn/wsjs/n429/n432/n1485/n1496/userobject1ai79957.html>>. [Access on 30 April 2015].
- [27] Sorensen P and Church R (2010). Integrating expected coverage and local reliability for emergency medical services location problems. *Socio-Economic Planning Sciences*, 44, pp. 8-18.
- [28] Statistics Department (2003). *The criteria of setting up emergency medical site in districts of Shanghai*. Shanghai.
- [29] Statistics Department (2009). *Report of occupational composition of employed population in Shanghai*. Shanghai.
- [30] Statistics Department (2011a). *The demographic yearbook in Changnin district*. Shanghai: Academia Press.
- [31] Statistics Department (2011b). *The demographic yearbook in Hongkou district*. Shanghai: Shanghai Academy of Social Sciences Press.
- [32] Statistics Department (2011c). *The demographic yearbook in Huangpu district*. Shanghai: Shanghai Cultural Publishing House.
- [33] Statistics Department (2011d). *The demographic yearbook in Jingan district*. Shanghai: Academia Press.
- [34] Statistics Department (2011e). *The demographic yearbook in Putuo district*. Shanghai: Shanghai Academy of Social Sciences Press.
- [35] Statistics Department (2011f). *The demographic yearbook in Xuhui district*. Shanghai: Academia Press.
- [36] Statistics Department (2011g). *The demographic yearbook in Yangpu district*. Shanghai: Shanghai Higher Education Electronic & Audio-visual Press.

- [37] Statistics Department (2011h). *The demographic yearbook in Zhabei district*. Shanghai: Shanghai Academy of Social Sciences Press.
- [38] Toregas C, Swain R, ReVelle C and Bergman L (1971). The location of emergency service facilities. *Operations Research*, 19 (6), pp. 1363-1373.
- [39] Tuzkaya UR, Heragu SS, Evans GW and Johnson M (2014). Designing a large-scale emergency logistics network - A case study for Kentucky. *European Journal of Industrial Engineering*, Volume 8, pp. 513-532.
- [40] Yin P and Mu L (2012). Modular capacitated maximal covering location problem for the optimal siting of emergency vehicles. *Applied Geography*, 34, pp. 247-254.
- [41] Zarandi MHF, Davari S and Sisakht SAH (2013). The large-scale dynamic maximal covering location problem. *Mathematical and Computer Modelling*, 57(3-4), pp. 710-719.

Appendix 1. Candidates for setting up emergency medical sites.

No.	Name of Location
1	Huajing Community Hospital
2	Shanghai Xuhui Changqiao Community Hospital
3	Shanghai Sports Hospital
4	Shanghai Xuhui Xinle Community Hospital
5	Shanghai Xuhui Maternal and Child Health-Care Center
6	Shanghai Xuhui Tianlin Community Hospital
7	The Sixth People's Hospital of Shanghai Jiaotong University
8	Wanping Hospital of Shanghai Xuhui District
9	The Hospital North Caoxi Road
10	Shanghai Changning Chengjiaqiao Community Hospital
11	Shanghai Gamma Knife Hospital
12	Tumour Hospital Affiliated to Fudan University
13	Shanghai Bokang Reproductivity Hospital
14	The Second Pulmonary Hospital of Shanghai
15	Shanghai Xujiashui District Hospital
16	Zunyi Hospital of Shanghai Changning District
17	International Welfare Association International Peace Maternity and Child Health Hospital
18	Branch of No.455 Hospital of the PLA
19	Rihui Hospital of Shanghai Xuhui District
20	Shanghai Xuhui Yixian Hospital
21	Zhongshan Hospital Affiliated to Fudan University
22	Shanghai Thoracic Hospital Affiliated to Shanghai Jiaotong University
23	Paediatric Hospital Affiliated to Fudan University
24	Shanghai Civil Aviation Hospital
25	Southeast Hospital of Shanghai Luwan District
26	Shanghai Ganghua Hospital
27	Shanghai Xuhui District City Tianping Road Lot Hospital
28	Shanghai Xuhui Dental Disease Prevention Center
29	No.455 Hospital of PLA
30	Shanghai Jiangnan Shipyard workers hospital
31	Shanghai Stomatological Medical Center
32	The Ninth People's Hospital Affiliated to Shanghai Jiaotong University School of Medicine
33	Shanghai Guanghua Hospital (Traditional Chinese and Western Medicine Hospital)
34	Shanghai Changning Xinhua Community Hospital
35	Shanghai Maternal and Child Health Hospital of Luwan District
36	Shanghai Dapujiao Lot Hospital
37	Shanghai Xuhui Yongjia Lot Hospital
38	Xianxia Hospital of Shanghai Changning District
39	The Third Hospital of Shanghai textile Shanghai Friendship Hospital

- 40 Otolaryngological Hospital Affiliated to Fudan University
- 41 Shanghai Second People's Hospital
- 42 Shanghai Huangpu Bansongyuan Community Hospital
- 43 Shanghai Oriental Breast Disease Hospital
- 44 Chinese Medicine Hospital of Shanghai Changning Tianshan District
- 45 Wuyi Hospital of Shanghai Changning District
- 46 Maternal and Child Health Hospital of Changning District
- 47 Shanghai Skin and Venereal Disease Prevention and Treatment Center
- 48 Shanghai Luwan Shunchang District Hospital
- 49 Obstetrics and Gynecology Hospital of Fudan University
- 50 Shanghai Electric Power Hospital
- 51 Shanghai Luwan Ruijin Erlu Community Hospital
- 52 Shanghai Yuanyang Hospital
- 53 Shanghai Xiangshan Hospital of Traditional Chinese Medicine
- 54 Shanghai Changning Beixinjing District Hospital
- 55 Shanghai Huangpu District Hospital of Traditional Chinese and Western Medicine
- 56 Central Huaihai Road Hospital of Luwan District
- 57 Huashan Hospital Affiliated to Fudan University
- 58 Central Hospital of Shanghai Xuhui District
- 59 Shanghai Changning Zhoujiaqiao Community Hospital
- 60 Luwan Branch of Ruijin Hospital (Center Hospital of Luwan District)
- 61 Shanghai City Huangpu District Dongjiadu Area Hospital
- 62 Shanghai Post and Telecommunications Hospital
- 63 Shanghai South City Maternal and Child Health Hospital
- 64 Shanghai First Maternal and Infant Health-Care Hospital
- 65 Tongren Hospital of Shanghai
- 66 Huayang Hospital of Shanghai Changning District
- 67 Yu Garden Hospital for the Elderly
- 68 Shuguang Hospital (West Part)
- 69 Shanghai Huangpu Xiaodongmen Lot Hospital
- 70 Shanghai West Nanjing Road Hospital
- 71 The children's Hospital Affiliated to Shanghai Jiaotong University
- 72 District Hospital of Jingan Temple
- 73 The Square Hospital of Huangpu District
- 74 Shanghai Hongguang Hospital
- 75 Workers' Hospital of Shanghai for Public Services
- 76 Shanghai Jingan Maternal and Child Health Care Centre
- 77 Shanghai Jingan Caojiadu Community Hospital
- 78 Geriatrics Hospital of Shanghai Jing'an District
- 79 Shanghai Huangpu Jinling Road Lot Hospital
- 80 Changzheng Hospital of Second Military Medical University in Shanghai
- 81 Renji Hospital of Shanghai Jiaotong University (West Campus)
- 82 Central Hospital of Shanghai Huangpu District
- 83 Shanghai Ophthalmological Hospital
- 84 Shanghai Occupation Disease Prevention and Treatment Institute
- 85 Shanghai Jingan Shimener Road Lot Hospital
- 86 Shanghai Huangpu East Nanjing Road Hospital
- 87 Shanghai Putuo Jade Street Hospital
- 88 The Branch of Huangpu Central Hospital (traditional Chinese Medicine)
- 89 Shanghai Jingan Jiangning Road Lot Hospital
- 90 The Bund Hospital at Huangpu District
- 91 Caoyang Red Cross Hospital
- 92 Shanghai Stomatosis Prevention and Treatment Center
- 93 Shanghai Haining Hospital
- 94 Haining Community Hospital of Zhabei District
- 95 Changzhen United Hospital of Putuo District
- 96 Shanghai Putuo District City Hospital of traditional Chinese Medicine
- 97 Shanghai Hongkou Zapu Community Hospital
- 98 Shanghai Seaman's Hospital
- 99 Shanghai Hongkou Tilanqiao Community Hospital
- 100 Shanghai Yangpu Pingliang Subdistrict Hospital
- 101 Shanghai Maternal and Child Health Hospital of Yangpu District
- 102 Shanghai Yangpu East Hospital (Shanghai Second Textile Hospital)
- 103 Shanghai Putuo Zhongshan North Subdistrict Hospital
- 104 The Center Hospital of Shanghai City Labor Bureau
- 105 Shanghai Hudong Nursing Centre
- 106 Shanghai Hangdao Hospital
- 107 Baoshan Hospital of Shanghai Zhabei District
- 108 Shanghai Yangpu District City Hospital of traditional Chinese Medicine
- 109 Shanghai Zhabei District City Zhi Jiangxi District Hospital
- 110 Jiaxing Hospital of Shanghai Hongkou District
- 111 Shanghai Hospital of Traditional Chinese Medicine
- 112 Shanghai Yichuan Hospital
- 113 Geriatric Hospital of Shanghai Yangpu District
- 114 Branch of Shanghai First People's Hospital (Fourth People's Hospital of Shanghai)
- 115 No.411 Hospital of the PLA
- 116 Shanghai Hongkou Ouyang Community Hospital
- 117 Shanghai Hongkou District Xingang Road Lot Hospital
- 118 Shanghai Yangpu Dentistry
- 119 Shanghai City Bridge Community Hospital

- 120 Ganquan Hospital of Shanghai Putuo District
- 121 Shanghai Zhabei District City Hospital of Traditional Chinese Medicine
- 122 Shanghai Yangpu Jiangpu Community Hospital
- 123 Gonghexin Community Hospital of Shanghai Zhabei District
- 124 Public Health Center affiliated to Fudan University (Shanghai Infectious Disease Hospital)
- 125 Shanghai Zhabei Women's Health Institute
- 126 Health Service Center of Siping Community (Siping Hospital of Shanghai Yangpu District)
- 127 Dinghai Hospital of Shanghai Yangpu District
- 128 Shanghai Yangpu Fengcheng District Hospital
- 129 Shanghai Hongkou Guangzhong District Hospital
- 130 Shanghai Putuo District Taopu Area Hospital
- 131 Wanrong Hospital of Shanghai Zhabei District
- 132 Kongjiang Hospital of Shanghai Yangpu District
- 133 The Hospital of Tongji University
- 134 Yanji Hospital of Shanghai Yangpu District
- 135 Yonghe Branch of Huashan Hospital Affiliated to Fudan University
- 136 Health Care Center, Medicine College at the Fudan University
- 137 Shanghai Hongkou Liangcheng Community Hospital
- 138 Shanghai Yangpu Wujiaochang Community Hospital
- 139 Shanghai Occupation Disease Hospital (Pulmonary Hospital of Tongji University)
- 140 Shanghai Jiangwan Hospital
- 141 Shanghai Zhabei Pengpu Community Hospital
- 142 Oriental Hepatobiliary Surgical Hospital of the Second Military Medical University
- 143 Shanghai Yangpu Minking Community Hospital
- 144 Shanghai Skin and Venereal Disease Hospital (Shanghai Zunyi Hospital)
- 145 Central Hospital of Shanghai Yangpu District of Central Plains Branch
- 146 Shanghai Yangpu Yinhang Community Hospital